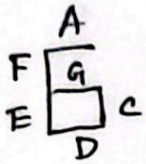
1. a) $\left(200 + \frac{60 \times x \times 10^9}{1440 \times 1024} + 300\right) \times \frac{3648}{x}$

$= 148456500$

$\therefore x = 96B$.

b) 

G, Not Connected, B

$\uparrow \quad \uparrow \quad \uparrow$

DDRC = 11110101

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

D C E A F

$0 \rightarrow$ Input $\quad 1 \rightarrow$ Output

2. a) i. ADD AX, BX

$$
\begin{array}{r}
3A53 \\
(+)\ 4D31 \\
\hline
8784
\end{array}
$$

(+) 0011

0100 (+)

1000 0111 1000 0100

(−)

PF = 1 → even 1's in lower order 8 bits

OF = 1 → (+) + (+) = (−)

ii. SUB AX, CX

$$
\begin{array}{r}
8784 \\
(-)\ 2BA1 \\
\hline
5BE3
\end{array}
$$

(−) 1000

0010 (+)

0101 1011 1110 0011

(+)

PF = 0 → odd 1's in lower order 8 bits

OF = 1 → (−) − (+) = (+)

b) Multiplexed pins are the connections which can serve multiple works. For 8086, multiplexed pins AD0 -AD15 are multiplexed pins as they can serve as A.B. and D.B.

3. a) We know,

P.A. = Segment * 10H + Offset

i) Now, for SS:
P.A. = 7CA2 * 10H + 3344 = 7FD64 H

Segment * 10 H = 7FD64 - B294 = 74AD0 H
Segment = (74AD0/10) = 74AD H (Ans.)

ii) Now, for DS:
P.A. = 6D2A * 10H + 11AA = 6E44A H

Offset = 6E44A - 2B41 * 10 H = 4303A H => 20 bits
This is invalid because here offset > 16 bits.

b) Let,

The A.B. of SRAM is x bits and D.B. of SRAM is y bits.
So, the A.B. of DRAM is 2x bits and D.B. of DRAM is 2y bits.

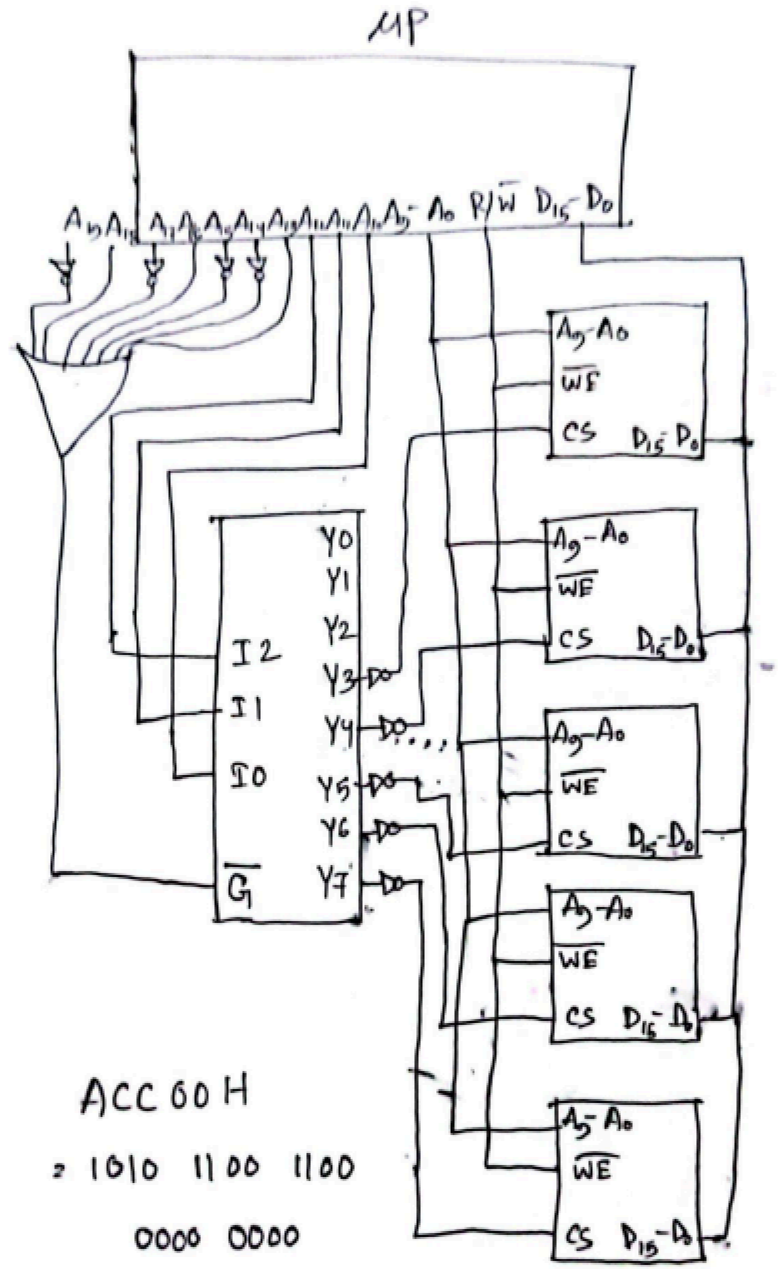
We know, $2^{A.B.}$ x D.B. = Ram Size
ATQ,
$2^x * y = 8 * 2^{10} * 8$ .....(i)
$2^{2x} * 2y = 64 * 2^{10} * 2^{10} * 8$ .....(ii)
Dividing (ii) by (i) we get,
$2^x * 2 = 8 * 2^{10} \Rightarrow 2^x = 2^{12}$ So, $x = 12$
For SRAM,
2^{12} x $y = 8 * 2^{10} * 8 \Rightarrow y = 16$
A.B. of SRAM = 12 bits
D.B. of DRAM = 2*16 = 32 bits

4. Ram capacity = $2^{10} * 16$ bits = 2 KB
No, of RAMs required = 10 KB / 2 KB = 5

Address mapping

2nd Ram:
3rd last address = AD3FD H
5th last address = AD3FB H
7th last address = AD3F9 H

4th Ram:
3rd last address = ADBFD H
5th last address = ADBFB H
7th last address = ADBF9 H



ACC00H

= 1010 1100 1100

0000 0000

5(a)

No. of interrupts = $\frac{2 \times 2^{10}}{4}$

$= 512$

last interrupt = 511

= 1FFH

lower byte of IP = 1FFH * 4H

= 007FC H

lower byte of CS = 007FEH

(Answer)

physical address 1C203H

IP = 2003 H

$$\therefore CS = \frac{1C203H - 2003H}{10H}$$

$$= 1A20H$$

Stack content

Sequence :

1. Push Flag register
2. Clear TF and IF
3. Push CS
4. Push IP
5. Load new IP and CS from IVT